References: Intro

Appendix

Section	Reference
jmap	The `jmap` command (Java SE 25)
jcmd	The `jcmd` command (Java SE 25)
jps	The `jps` command (Java SE 25)
jstat	The `jstat` command (Java SE 25)
jmap, jcmd, jps, jstat	Troubleshooting Guide: Diagnostic Tools (Java SE 25)
JOL	See References: JOL, the next slide

References: JOL

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Topic	Reference
JOL	OpenJDK Code Tools: Java Object Layout (JOL)
Source, CLI	openjdk/jol: jol-core, jol-cli and jol-samples
Samples	jol-samples: headers, alignment, arrays and more
Library	Maven Central: `org.openjdk.jol:jol-core`
Headers	JEP 450: Compact Object Headers (Experimental), JDK 24
Headers	JEP 519: Compact Object Headers, JDK 25
Headers	OpenJDK Project Lilliput
Flag	The `java` command (Java SE 25): `-XX:+UseCompactObjectHeaders`

References: Baseline and volatile heap

Appendix

- HotSpot Virtual Machine Garbage Collection Tuning Guide (Java SE 25)
- Factors Affecting Garbage Collection Performance (Java SE 25)
- The `java` command: `-Xmx` and GC options (Java SE 25)

References: jmap in the wild

Appendix

- The `jmap` command (Java SE 25)
- The `jps` command (Java SE 25)
- Troubleshooting Guide: Diagnostic Tools (Java SE 25)

References: Movers

Appendix

- Java Object Layout (JOL), OpenJDK Code Tools
- `Integer.valueOf(int)` (Java SE 25 API)
- `HashMap` (Java SE 25 API)
- `ArrayList` (Java SE 25 API)
- The `java` command: `-XX:+UseCompactObjectHeaders` (Java SE 25)

References: Shakers

Appendix

- Package `java.util.stream` (Java SE 25 API)
- The Java Tutorials: Lambda Expressions
- Java HotSpot VM Performance Enhancements: Escape Analysis (Java SE 25)
- Java Magazine: Escape analysis in the HotSpot JIT compiler
- HotSpot Virtual Machine Garbage Collection Tuning Guide (Java SE 25)

References: The problem

Appendix

- HotSpot Virtual Machine Garbage Collection Tuning Guide (Java SE 25)
- Package `java.util.stream` (Java SE 25 API)
- The `jmap` command (Java SE 25)
- Java Object Layout (JOL), OpenJDK Code Tools

References: "Dollar" classes

Appendix

- The Java Tutorials: Anonymous Classes
- JLS SE 25, 15.9.5: Anonymous Class Declarations
- JLS SE 25, 13.1: The Form of a Binary (`$` names)
- JLS SE 25, 8.1.3: Inner Classes and Enclosing Instances

References: Object pooling

Appendix

- `Integer.valueOf(int)` (Java SE 25 API): the -128 to 127 cache
- JLS SE 25, 5.1.7: Boxing Conversion
- The Java Tutorials: Thread Pools
- Java Garbage Collection Basics: why short-lived objects are cheap

References: Compressed OOPs

Appendix

- Java HotSpot VM Performance Enhancements (Java SE 25): Compressed Ordinary Object Pointer
- The `java` command (Java SE 25): `-XX:+UseCompressedOops`

References: Compact object headers

Appendix

- JDK 25 Release Notes: JEP 519 Compact Object Headers
- The Arrival of Java 25 (Oracle Java blog)
- The `java` command (Java SE 25)

References: Stream API

Appendix

- `java.util.stream` package summary (Java SE 25): pipelines, laziness, low-level construction
- `Stream` (Java SE 25 API)
- The Java Tutorials: Aggregate Operations
- The Java Tutorials: Reduction

References: Lambdas

Appendix

- The Java Tutorials: Lambda Expressions
- JLS SE 25, 15.27: Lambda Expressions (15.27.2 effectively final, 15.27.4 evaluation)
- `LambdaMetafactory` (Java SE 25 API)
- Java 8: Lambdas, Part 1 (Oracle technical article)

References: Escape analysis

Appendix

- Java HotSpot VM Performance Enhancements (Java SE 25): Escape Analysis
- Escape Analysis in the HotSpot JIT Compiler (Oracle Java Magazine)
- The `java` command (Java SE 25): `-XX:+DoEscapeAnalysis`

References: Solutions

Appendix

- OpenJDK Code Tools: Java Microbenchmark Harness (JMH)
- OpenJDK Code Tools: Java Object Layout (JOL)
- JEP 519: Compact Object Headers
- The `jcmd` command (Java SE 25)
- The Java Tutorials: Thread Pools
- Eclipse Collections